

P8X Backplane Bus Definition

Revision C — June 2026 — FOR REVIEW

Connector: DIN 41612, 3 rows (A/B/C) × 32 pins. Backplane: 10 slots, 25.4 mm pitch, 4-layer (signals / GND plane / +5V plane / signals). Power: A1,B1,C1,A2,B2,C2 = +5V; A31,B31,C31,A32,B32,C32 = GND. Row B pins 3–30 are a grounded guard row between the two signal rows. This table is generated from the same pin-map function used by the CAD generators, so it cannot drift from the board files.

Pin Assignment (component-side view of card connector)

Pin	Row A	Row B	Row C
1	+5V	+5V	+5V
2	+5V	+5V	+5V
3	D0	GND	A0
4	D1	SPARE12	A1
5	D2	GND	A2
6	D3	SPARE13	A3
7	D4	GND	A4
8	D5	SPARE14	A5
9	D6	GND	A6
10	D7	SPARE15	A7
11	-RES	GND	A8
12	DOE0	SPARE16	A9
13	DOE1	GND	A10
14	DOE2	SPARE17	A11
15	DOE3	GND	A12
16	DLD0	SPARE18	A13
17	DLD1	GND	A14
18	DLD2	SPARE19	A15
19	DLD3	GND	ALUS0
20	PSEL0	SPARE20	ALUS1
21	PSEL1	GND	ALUS2
22	PINC	SPARE21	ALUS3
23	PDEC	GND	ALUM
24	CLK	SPARE22	CIN
25	CLKB	GND	SH0
26	LDF	SPARE23	SH1
27	FC	CLRC	PSEL2
28	FZ	BSEL	LDZN
29	FN	IRQ	SHCIN
30	FV	SPARE11	SETC
31	GND	GND	GND
32	GND	GND	GND

Shading: red = +5V, blue = GND, grey = spare (bused).

Legend

Signal	Acronym	Meaning		
Address bus, bit n (8-bit)	An	Address bus, bit n (16-bit)		
Output Enable field: selects which card drives the data bus	DLD0-3	Data Load field: selects which register latches the data bus		
Pointer SElect: which of P0-P3 (ptr/ptr/SP) is active	PINC / PDEC	Pointer INCrement / DECrement (applies to selected pointer)		
ALU function Select lines	ALUM	74181 Mode: logic vs arithmetic		
Carry IN	SH0-1	SHifter control (pass/left/right/rotate)		
Flags: latch C,Z,N,V from ALU	-RES	RESet, active low (_N = active low)		
System CLock and its complement (CLK-Bar); loads on rising CLK, strobes gated by CLKB	SPAREn	Unassigned, bused to all 10 slots, reserved for future use	FC FZ FN FV	ALU flags to control (condition mux), also former SPARE0-3

Signal Dictionary

Signal	Pin(s)	Dir*	Description
D0	A3	<>	Data bus bit 0
D1	A4	<>	Data bus bit 1
D2	A5	<>	Data bus bit 2
D3	A6	<>	Data bus bit 3
D4	A7	<>	Data bus bit 4
D5	A8	<>	Data bus bit 5
D6	A9	<>	Data bus bit 6
D7	A10	<>	Data bus bit 7
A0	C3	RB>	Address bus bit 0
A1	C4	RB>	Address bus bit 1
A2	C5	RB>	Address bus bit 2
A3	C6	RB>	Address bus bit 3
A4	C7	RB>	Address bus bit 4
A5	C8	RB>	Address bus bit 5
A6	C9	RB>	Address bus bit 6
A7	C10	RB>	Address bus bit 7
A8	C11	RB>	Address bus bit 8
A9	C12	RB>	Address bus bit 9
A10	C13	RB>	Address bus bit 10
A11	C14	RB>	Address bus bit 11
A12	C15	RB>	Address bus bit 12
A13	C16	RB>	Address bus bit 13
A14	C17	RB>	Address bus bit 14
A15	C18	RB>	Address bus bit 15
DOE0	A12	<>	Data bus bit OE0
DOE1	A13	<>	Data bus bit OE1
DOE2	A14	<>	Data bus bit OE2
DOE3	A15	<>	Data bus bit OE3
DLD0	A16	<>	Data bus bit LD0
DLD1	A17	<>	Data bus bit LD1
DLD2	A18	<>	Data bus bit LD2
DLD3	A19	<>	Data bus bit LD3
PSEL0	A20	C>	Pointer select bit 0
PSEL1	A21	C>	Pointer select bit 1
PSEL2	C27	C>	Pointer-select bit 2 (rev B: P0-P3 + PT scratch=4); to reg-bank
PINC	A22	C>	Selected pointer increment
PDEC	A23	C>	Selected pointer decrement
LDF	A26	C>	Latch flags (ALU card)
ALUS0	C19	C>	74181 function select S0
ALUS1	C20	C>	74181 function select S1
ALUS2	C21	C>	74181 function select S2

Signal	Pin(s)	Dir*	Description
ALUS3	C22	C>	74181 function select S3
ALUM	C23	C>	74181 mode (logic/arith)
CIN	C24	C>	ALU carry in
SH0	C25	C>	Shifter control 0
SH1	C26	C>	Shifter control 1
CLK	A24	C>	System clock
CLKB	A25	C>	Inverted clock
-RES	A11	C>	System reset, active low
LDZN	C28	C>	Latch Z,N from the data bus on loads (rev B); to ALU card
SHCIN	C29	C>	Shifter shift-in = C flag, for rotate-through-carry (rev B); to ALU card
SETC	C30	C>	Force C flag = 1 (SEC, rev B); to ALU card
CLRC	B27	C>	Force C flag = 0 (CLC, rev B); to ALU card
BSEL	B28	C>	ALU B-input mux select (rev C): 0=B register, 1=T register; to ALU card
FC	A27	ALU>	Flag: Carry. Driven continuously by the ALU card flag register; read by the control card
FZ	A28	ALU>	Flag: Zero. ALU card to control card, as FC
FN	A29	ALU>	Flag: Negative (result bit 7). ALU card to control card, as FC
FV	A30	ALU>	Flag: oVerflow. ALU card to control card; rev A ALU card drives 0 (V unimplemented, 1=V)
IRQ	B29	C>	Maskable interrupt request (rev C, reserved). Devices pull it; control-card interrupt control
SPARE11	B30	n/a	Reserved, bused across all slots
+5V	A1, A2, B1, B2, C1, C2	PWR	Power, 6 pins total
GND	A31 ... C32 (18 pins)	PWR	Ground / row-B guard

Dir Field Legend

Symbol	Meaning
C>	Driven by the Control card; received by all other cards
RB>	Driven by the Register Bank card (sole address-bus driver)
ALU>	Driven by the ALU card (flag register outputs)
<>	Bidirectional: exactly one driver per microcycle, enforced by one-hot DOE decode
PWR	Power distribution pin (+5V or GND planes)
n/a	Spare: bused across all slots, no driver assigned

DOE / DLD Field Encodings

Code	DOE: bus source	DLD: bus destination
0	(bus idle - pulled up on backplane)	none
1	A register	A register
2	B register	B register
3	T (hidden temp)	T (hidden temp)
4	T2 (hidden temp)	T2 (hidden temp)
5	ALU result via shifter	FLAGS (restore)
6	FLAGS	IR (instruction register)
7	MEM read (memory / I/O)	MEMW write strobe (memory / I/O)
8	PTRL - selected pointer low byte	PTRL - load selected pointer low
9	PTRH - selected pointer high byte	PTRH - load selected pointer high
10-15	reserved	reserved

Review Notes

1. RC clock termination (backplane far end) ships DNP; populate only if scope shows ringing.
2. D0-D7 carry 10k pull-ups on the backplane only; cards never add bus conditioning.
3. Loads occur on rising CLK; write strobes are gated with CLKB (second half-cycle).
4. Address bus is driven exclusively by the register-bank card (selected pointer).
5. rev C3: C27-C30 = PSEL2/LDZN/SHCIN/SETC and B27 = CLRC (were SPARE4-8). rev C: B28 = BSEL (ALU B-mux select, was SPARE9), B29 = IRQ (interrupt request, was SPARE10); SPARE11 (B30) remains reserved.
- 5b. SPARE0-3 were reallocated as flag lines FC/FZ/FN/FV (A27-A30). SPARE numbering therefore starts at 4.
- 5c. Row B ground guard spans B3-B26; B27 = CLRC, B28 = BSEL, B29 = IRQ, B30 = SPARE11.
6. Verify row A/C orientation against physical DIN connectors before first fab.